Sunilraj D

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Education

Amrita Vishwa Vidyapeetham Chennai, India
B.Tech – Computer Science & Engineering (Artificial Intelligence) *2023 – 2027*

- CGPA: 7.64 / 10

Skills

Programming Languages: Python, Java

Machine Learning: Supervised and Unsupervised Learning, Classification, Clustering, Ensemble Learning, Concept Drift Detection, Fraud Detection, Model Evaluation

Deep Learning: CNN, U-Net, LSTM, BERT, Graph Neural Networks (GraphSAGE, GAT)

Frameworks: PyTorch, TensorFlow, Scikit-learn, PyTorch Geometric, Streamlit

Libraries: NumPy, Pandas, OpenCV, SciPy, Matplotlib, Seaborn, Plotly, XGBoost

Tools: Git, Linux, Jupyter Notebook, LaTeX

Certifications & Achievements

NPTEL: Python for Data Science – **Top 2%**, Score 83% | C Programming – Score 78% | **Coursera:** DSA | **Teachnook & IIT Roorkee:** Data Science

Experience

Warehouse Intelligence Systems Intern Chennai, India
TAFE – Tractors and Farm Equipment Limited *Feb 2026 – May 2026*

- Developed a five-module warehouse management system using Python and Streamlit for inventory, scoring, recommendations, analytics, and alerts, replacing manual notebook-based part recording in the engineering stores department.
- Designed a weighted scoring system to prioritize warehouse parts based on usage frequency and inactivity, enabling automated shelf placement recommendations.
- Built a K-Means clustering module ($k = 3$) to classify parts into Fast/Slow/Rare categories from usage logs. Integrated a stock health dashboard with low-stock alerts and Plotly analytics.

Projects

Hybrid CNN for Alzheimer's MRI Classification *Python* | *TensorFlow* | *CNN* | *Scikit-learn* | *XGBoost*
GitHub Repository

- Developed a hybrid CNN and stacking ensemble for four-class Alzheimer's MRI classification, achieving over 90.8% test accuracy.
- Co-authored an IEEE conference paper and contributed to model development, training, and performance evaluation.

Attention-Enhanced UNet for Lung Tumor Segmentation *Python* | *PyTorch* | *UNet* | *Medical Image Segmentation*
GitHub Repository

- Developed an Attention-Enhanced UNet model for lung tumor segmentation on CT scans, evaluating segmentation performance using Dice and IoU metrics.
- Applied customized loss functions and feature refinement techniques to address class imbalance and improve localization.

Zero-Shot EV Energy Prediction via NLP Knowledge Transfer *Python* | *BERT* | *XGBoost* | *LSTM* | *NLP*
GitHub Repository

- Developed a zero-shot framework to estimate EV energy consumption for unseen vehicle models using information extracted from owner manuals, without requiring historical driving data.
- Extracted semantic features from EV owner manuals using BERT embeddings and trained LSTM and XGBoost models to estimate energy consumption for previously unseen vehicle models.

Drift-Aware Adaptive Fraud Detection *Python* | *XGBoost* | *PyTorch Geometric* | *GraphSAGE* | *GAT*
GitHub Repository

- Developed a concept drift detection and adaptation framework for evolving financial transaction streams using PSI, Maximum Mean Discrepancy (MMD), and Page-Hinkley on the Elliptic Bitcoin transaction dataset.
- Compared static, sliding-window, and drift-triggered retraining strategies to evaluate post-drift fraud detection performance and retraining efficiency.
- Implemented GraphSAGE–XGBoost and simplified time-aware graph-attention–XGBoost hybrids; evaluated robustness using multiple random seeds and statistical significance tests.